BotLog TypeScript Validator - Tier 1 $500 Bounty Implementation

Overview

This PR implements the **Log Validator in TypeScript/JavaScript** as described in the README Tier 1 $500 bounty.

Features Implemented

- ✅ **Ed25519 Signing** using `@noble/ed25519` (same as Python ref impl uses PyNaCl)
- ✅ **SHA-256 Hashing** using `@noble/hashes` (built-in equivalent)
- ✅ **RFC 8785 Canonical JSON** - deterministic key sorting for verifiable hashing
- ✅ **Single Entry Verification** - signature + log_hash validation
- ✅ **Chain Verification** - hash chain integrity + timestamp monotonicity per actor
- ✅ **Tamper Detection** - detects broken chain, fake previous_hash, timestamp regression

Structure

```
reference/typescript/
├── botlog.ts       # Core validator (BotLogEntry class + verifyChain)
├── example.ts      # 3-entry chain demo (human -> human -> ai)
├── test.ts         # 4 unit tests
├── package.json    # Dependencies
└── README.md       # This file
```

How to Test

```bash
cd reference/typescript
npm install
npm run test  # 4 tests should pass
npm run demo  # Chain verification: PASSED
```

Compliance with Spec

Matches `reference/python/botlog-mini/botlog.py` behavior:

- Same action types: propose|commit|execute|verify|dispute
- Same actor types: human|ai
- Same hashing: previous_hash + signature + canonical payload
- Same verification rules from README

Example Usage

```typescript
import { BotLogEntry, generateKeypair, publicKeyToBase64, verifyChain

const { privateKey, publicKey } = await generateKeypair();
const pubKeyB64 = publicKeyToBase64(publicKey);

const entry = new BotLogEntry({
  version: '1.0',
  timestamp: getCurrentTimestamp(),
  actor: { type: 'human', id: 'KullAxel', public_key: pubKeyB64 },
  action: { type: 'propose', description: 'Launch campaign' },
  previous_hash: null,
});
await entry.sign(privateKey);
console.log(entry.toJSON());
```

Bounty Claim

Claiming Tier 1 ($500) - Log Validator in TypeScript/JavaScript

Payout via Polar.sh or GitHub Sponsors as mentioned in README.
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Ready for review! 🚀